

Design a removable spill-mat insert

Learning intention

Plan a fair material comparison, record water-retention results and justify a product choice.

Brief: retain at least 6 g of water in every trial.

Choose the lowest-cost product that meets the brief.

A good decision needs a fair comparison.

Connect properties to a purpose

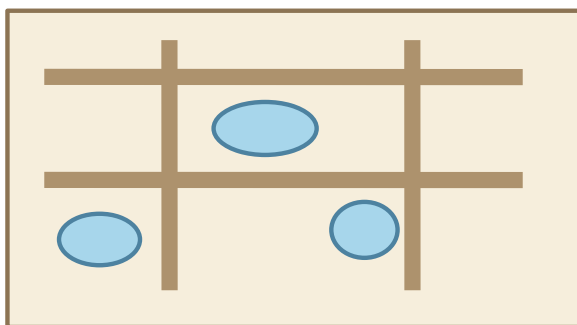
R1. What does absorbent mean?

R2. Which must stay the same: soak time or product type?

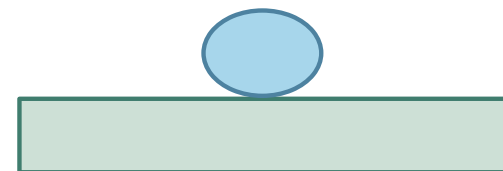
R3. Is a heavy wet cloth necessarily holding more water?

What are we actually testing?

Absorbing water is a useful material property



Fibres and connected spaces can take in and retain water.



A drop on an intact plastic sheet is not absorbed.

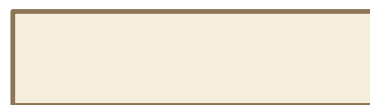
Close-up model; water marks and spaces are enlarged.

Absorbent structures take water in. Our measure is water held after a fixed drip time. Water sitting on a surface is not the same thing.

Subtract the starting mass

Constructed worked example: subtract dry mass

Dry sample



2.0 g

Wet after dripping



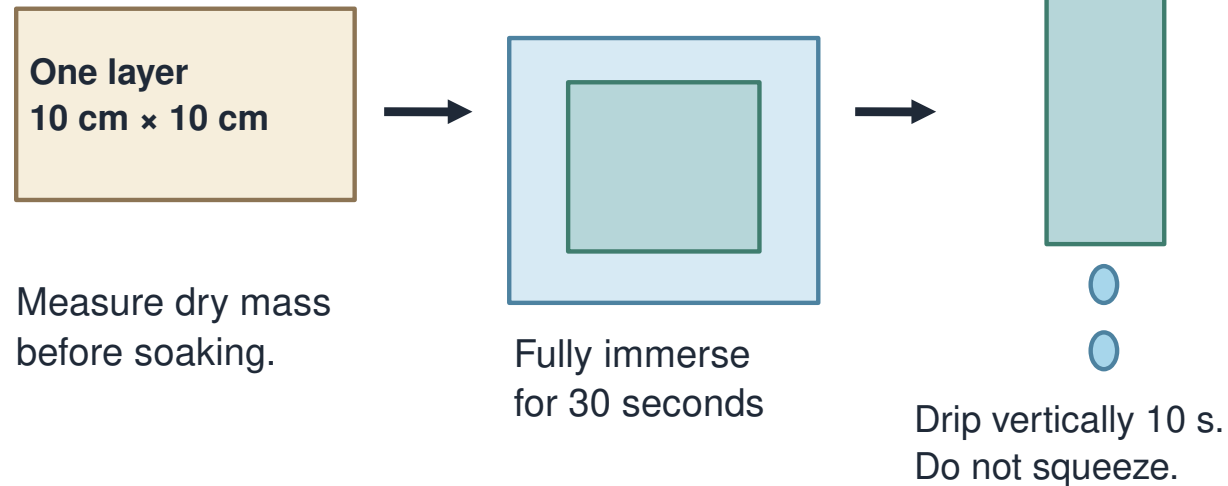
9.0 g

$$\text{Water retained} = 9.0 - 2.0 = 7.0 \text{ g}$$

Constructed example: dry sample 2.0 g. Wet sample after dripping: 9.0 g. Retained water: $9.0 - 2.0 = 7.0$ g.

Keep the question and method matched

Same method for each product sample



Use fresh dry samples for repeats. Then measure wet mass.

Change the product; measure retained water. Keep area, layers, soak time and drip time the same. Use fresh dry samples for three trials.

Rescue this unfair test plan

“Large felt: soak 60 s, squeeze hard.”

“Small paper: soak 10 s, weigh immediately.”

Use page 1 to find and repair the unfair changes.

Read a repeat, not a promise

Constructed sample data: retained water after the fixed method. The requirement is at least 6 g on every trial. Which products meet it in these three trials?

Product	Trial 1 / g	Trial 2 / g	Trial 3 / g
Kitchen paper	5.2	5.5	5.3
Cotton cloth	7.0	6.8	7.2
Felt sheet	8.0	7.8	8.2

Constructed sample results; one-layer 10 cm squares; fresh dry samples.

Best for which job?

Fictional costs per square: paper 2, cotton 4, felt 6 credits.

The brief asks for the lowest cost that passes every trial.

Choose and defend; then name one untested property.

What makes the comparison fair?

Which plan can answer which product retains more water?

- A** Use different sizes because products differ.
- B** Use equal-area dry samples and one fixed method.
- C** Reuse each wet sample for every repeat.
- D** Squeeze only the cotton before weighing.

Write the product team's report

Choose one route, with the common evidence page.

Plan the comparison and record retained water.

Recommend a product using the exact brief.

Record the result you have

Retained water = wet mass – dry mass.

Keep the unit g with each result.

Actual practical result? Record it honestly and separately.

Keep the conclusion within the evidence

Does the product pass every trial?

Is it the cheapest product that passes?

Which feature of a real mat still needs testing?

Challenge the design decision

One pupil recommends felt because it holds the most.

Another recommends cotton because it meets the cheaper brief.

Which claim answers our actual question?

Plan our next improvement

Which untested property should the team investigate?

Suggest a measurable outcome and one control.

Teacher: explain how one suggestion will shape the next test.

Exit: the fair-test check

- E1. Name what changes and what is measured.
- E2. Name two conditions that stay the same.
- E3. Recommend a product using one result and the brief.